Biochemistry Cumulative Examination

OCTOBER 5, 1996

You are asked to explain to a class of high school students the roles of hemoglobin and myoglobin in the delivery of oxygen to tissues in a mammal. Try to make your answers as accurate and clear as possible.

1. (10 points) Why are two O_2-binding proteins present in organisms?

2. (30 points) Discuss in detail the quaternary structures of hemoglobin and myoglobin, and the significance of quaternary structure for the functions of these two proteins. Are their quaternary structures different? If so, how are they different? Are these differences important for function? If so, how and why?

3. (30 points) Sketch approximate oxygen-binding curves for the two proteins. The protein of which quaternary structure has the higher oxygen affinity? Why? Which structure more readily releases oxygen? Why? Why are these questions significant for physiology?

4. (15 points) During vigorous exercise, acid may accumulate in the muscles. Why? Discuss a mechanism by which hemoglobin helps to remove acid from the muscles.

5. (15 points) The binding affinity for carbon monoxide (CO) to hemoglobin is about 200 times as great as that for oxygen. In 1935, Haldane and Priestly wrote that "a person whose blood [hemoglobin] is half-saturated with CO is practically helpless . . . but a person whose hemoglobin percentage is simply diminished to half by anemia may be going about work as usual." Suggest a molecular explanation for this.